Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: LedgerLens
b. Team leader name: Changhyeon (SK Telecom, AI CIC)
c. Problem Statement: BFSI: Intelligent Risk, Fraud & Financial Experiences

Why this matters

Small businesses run Southeast Asia's economy but cannot get a loan, because nobody can afford to read their records

They run the economy

64M

MSMEs in Indonesia alone

97% of jobs, 60% of GDP

Bank Indonesia, Aug 2026

But they cannot borrow

Vietnam: share with a bank loan

47.1% large firms

8.8% SMEs

70% of emerging-market MSMEs lack adequate financing

FiinGroup, Aug 2026 · IFC

Because banks cannot read them

75.5% have no acceptable collateral

VCCI Private Economic Report 2025

Records are PDFs, screenshots, receipts

no credit file, nothing a scorecard can use

Reading them costs more than the loan earns

high cost, low return for the lender (ADB 2025)

So they are

Declined,

or lent blind at high rates

PH: 3.9% of bank credit (ADB)

The opening

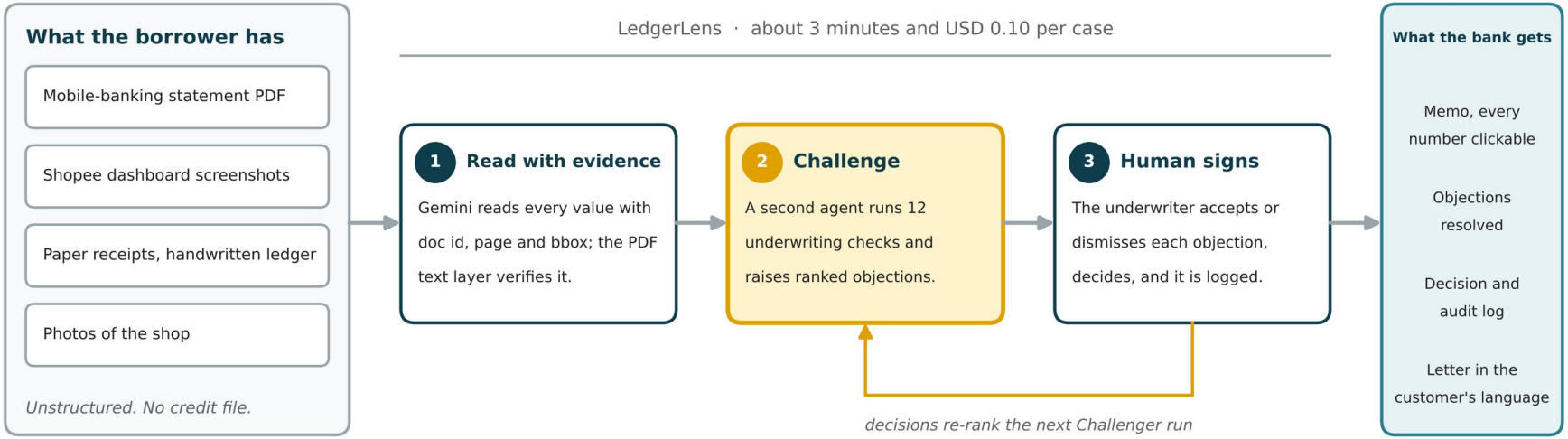
VPBank, Vietcombank and Agribank are moving from collateral to cash-flow lending (VTV, May 2026).

Cash-flow lending means someone has to read the cash flow. That reading is what LedgerLens makes fast, sourced and auditable.

Sources: IFC MSME Finance · FiinGroup via Viet Nam News, Aug 2026 · ADB Asia SME Monitor 2025 · Bank Indonesia, Aug 2026 · VCCI Private Economic Report 2025 and VTV, May 2026

Brief about the idea

LedgerLens turns the shoebox into a credit memo a bank can trust



In one sentence: *the first credit memo where every number can be clicked back to its source, and a second AI tries to break it before a human signs.*

Opportunities

What changes for the lender: the same documents, read in 3 minutes, every number defensible

	Collect	Read	Assess	Decide	Explain
Today	Same PDFs, screenshots, receipts	45 min by hand, retyped into Excel	Collateral and gut feel	Decline, or price blind at high rates	No reason the customer can read
With LedgerLens	Same documents, uploaded as one case	3 min, every figure linked to its line	Challenger raises 12 checks as objections	Underwriter signs with evidence in hand	Letter in the customer's language

45 → 3
minutes per file

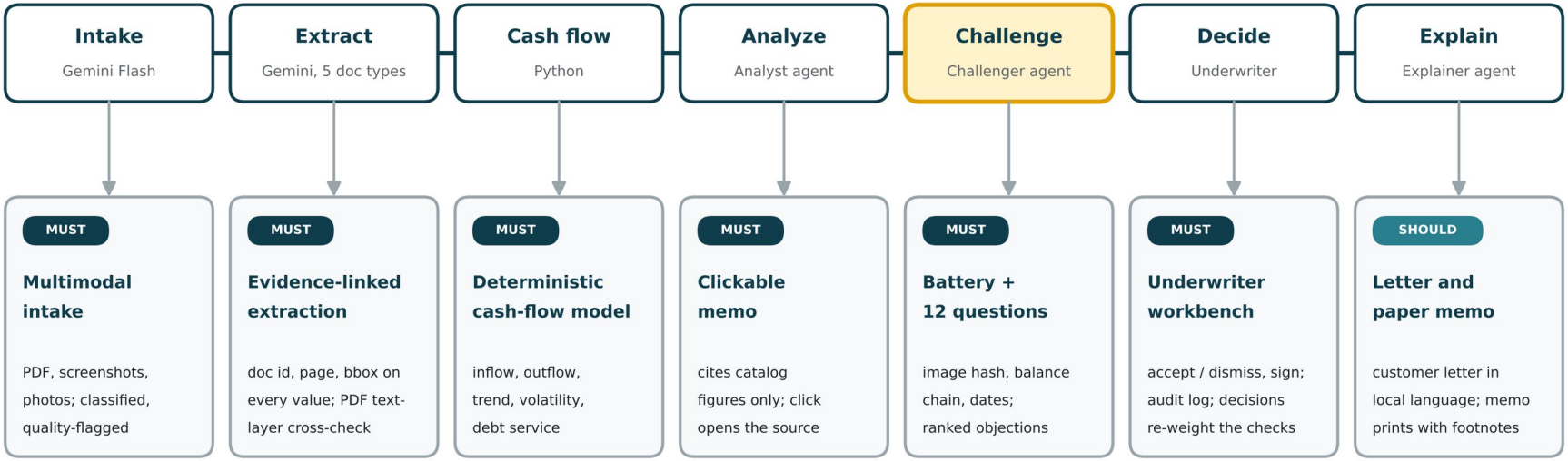
11 → 0.10
USD per case

0
unsourced numbers

Why it is different: the memo is written by one agent and attacked by another, every figure is computed by code and linked to its source, and the underwriter's decisions re-rank the next run. A memo, not a score.

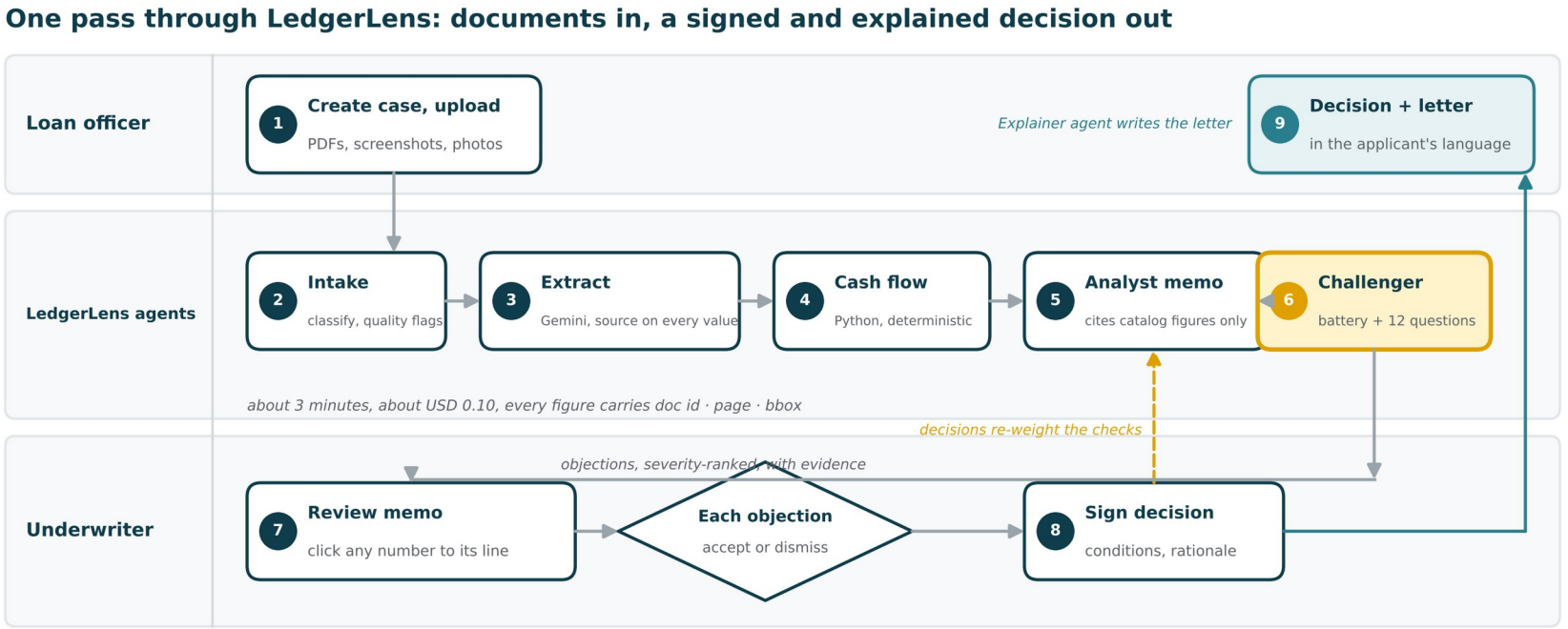
List of features offered by the solution

Every feature hangs on one pipeline, and the pipeline ends with a person



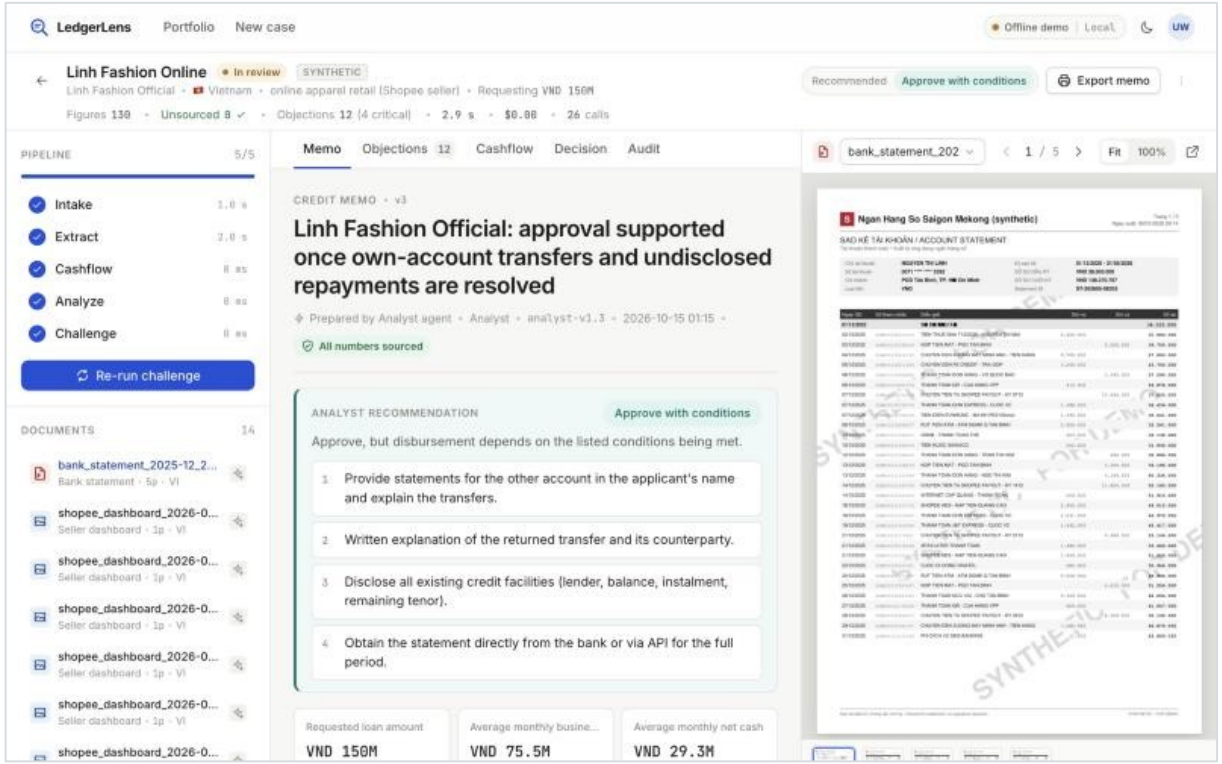
Out of scope by design: credit scoring, loan pricing, core-banking integration. LedgerLens informs the decision; it does not make it.

Process flow diagram



Three lanes: the loan officer uploads, the agents read and challenge, the underwriter decides. Decisions flow back into the Challenger's priorities.

Wireframes / Mock diagrams of the proposed solution



Live prototype, case B: 14 documents, 130 figures, 0 unsourced, 12 objections.

1 Memo with a recommendation

Headline, conditions and every figure drafted by the Analyst, from catalog figures only

2 Underlined numbers are evidence

Click one: the source page opens with the bounding box highlighted

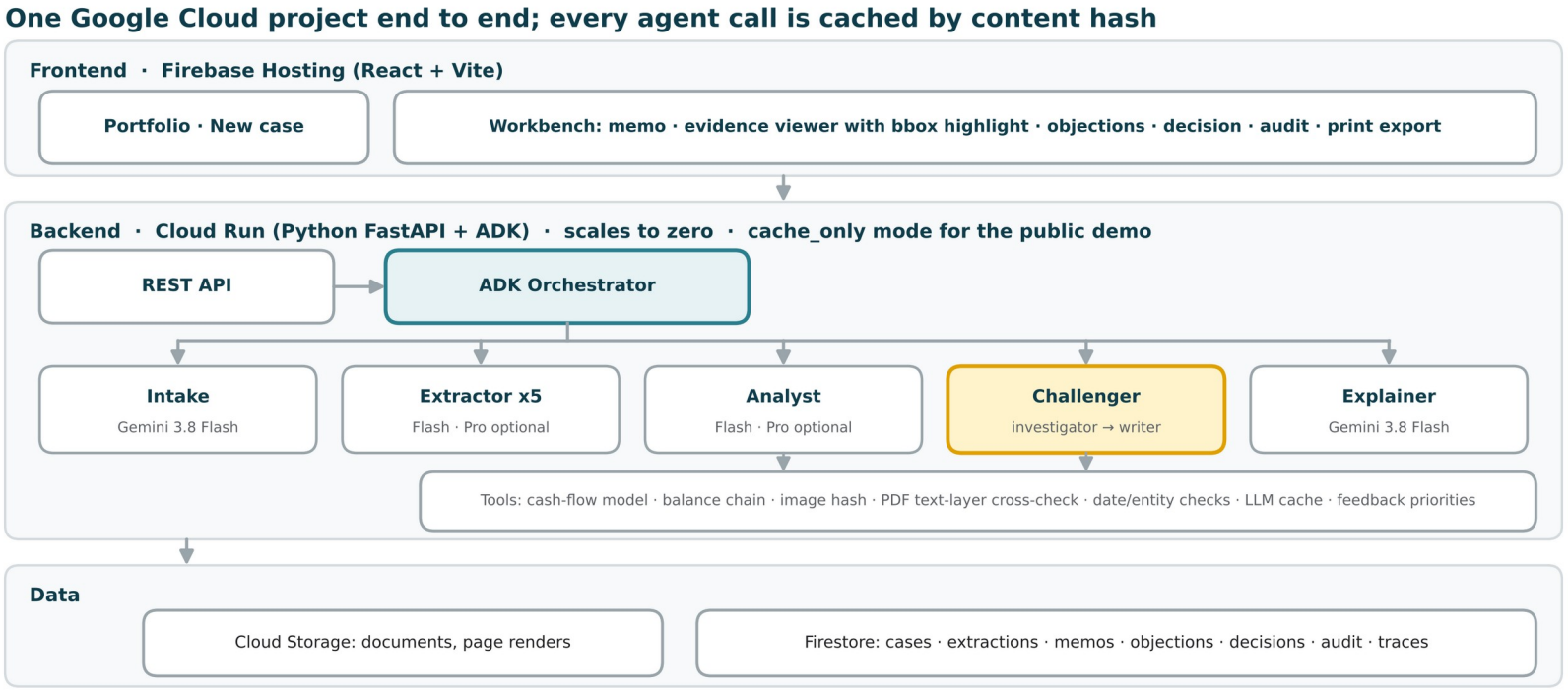
3 Unsourced 0, in the header

Enforced by the output schema; a miss becomes a Challenger objection

4 Pipeline, cost, calls

Five stages with timings; USD and LLM-call counters per case; Export memo prints a paper memo

Architecture diagram of the proposed solution



One Google Cloud project end to end. Every agent call is cached by content hash, so re-runs are free and deterministic.

Technologies to be used in the solution

One Google Cloud project, from the underwriter's screen down to the cache

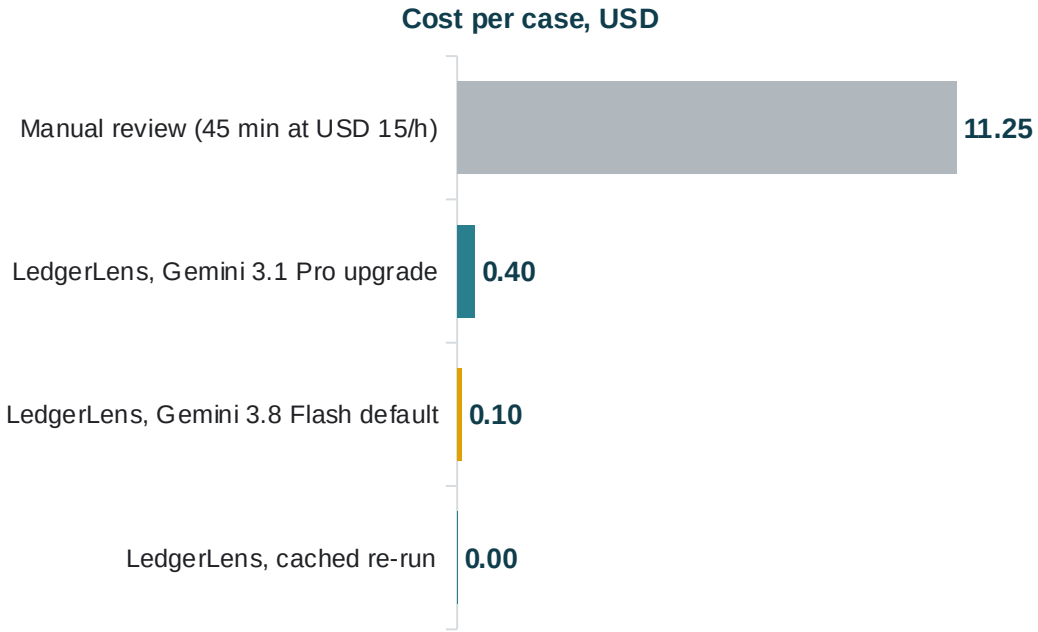
	What	Why this choice
Experience	Firebase Hosting · React + Vite: portfolio, new case, workbench, evidence viewer, print export	*One screen per decision; dark and light themes*
Agents	ADK: Intake · Extractor x5 · Analyst · Challenger (investigator, writer) · Explainer	*Challenger never sees the Analyst prompt*
Models	Gemini 3.8 Flash for every agent · Gemini 3.1 Pro optional for the heavy agents	*One env var switches to Pro*
Tools	Python: cash-flow model · balance chain · image hash · PDF text-layer cross-check · date/entity checks	*Numbers computed, never generated; the text layer verifies what Gemini read*
Data	Firestore: cases, extractions, memos, objections, decisions, audit, traces · Cloud Storage: documents	*Accept/dismiss rates become Challenger priorities*
Run	Cloud Run, scales to zero · content-hash LLM cache · cache_only demo · budget cap	*Re-runs and evaluation cost nothing*

Google Cloud Open source

Output schema requires source {doc_id, page, bbox} on every value. LEDGERLENS_MODEL_PRO=gemini-3.1-pro-preview switches the heavy agents to Pro.

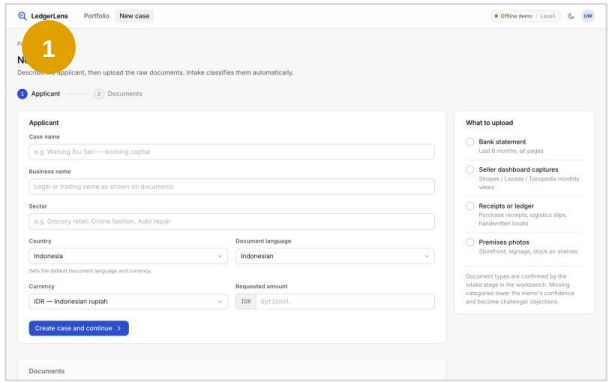
Estimated implementation cost

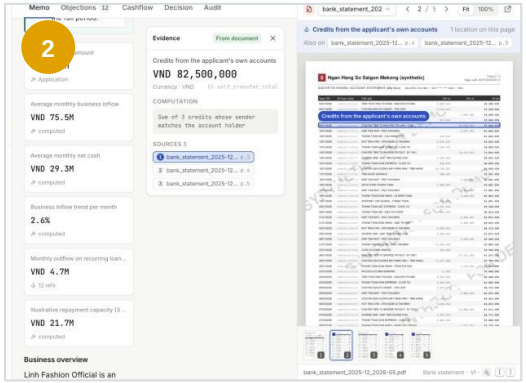
Cheap enough for small loans: about USD 0.10 per case on Flash, zero on a re-run

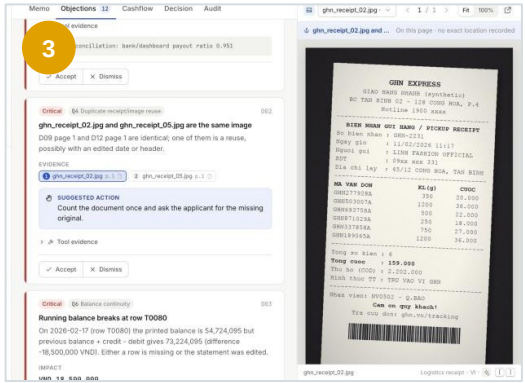


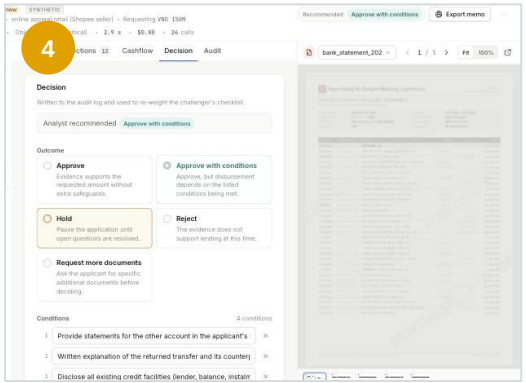
Assumption	Value
Gemini 3.8 Flash list price	USD 0.75 / 3.75 per 1M tokens
Gemini 3.1 Pro list price	USD 2.00 / 12.00 per 1M tokens
Tokens per 20-document case	60-90k input, 15-25k output
Hosting	Cloud Run scales to zero; Firestore and Storage in free tier
Guardrail	Budget-cap env var stops Gemini calls once spent
Build	4 engineers, 4 days, free tier and trial credits

Snapshots of the prototype









1 New case

Applicant details, then the raw documents. Intake classifies and quality-flags them.

2 Click a number, see the line

Evidence panel with computation and all sources; the statement opens with the bounding box highlighted.

3 Challenger objections

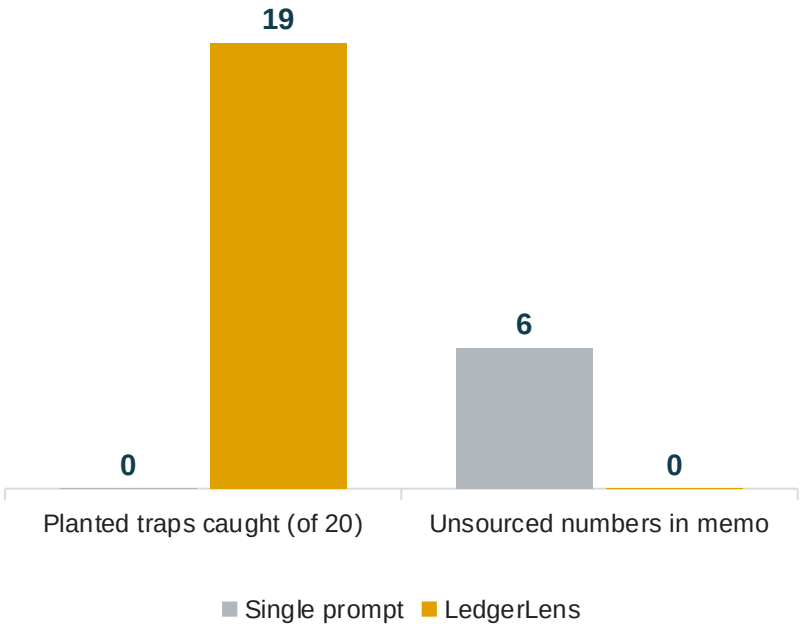
Critical: two receipt photos are the same image with an edited date. Evidence chips open the photo.

4 The underwriter signs

Outcome, conditions and rationale, written to the audit log; accept/dismiss rates re-weight the checks.

Prototype performance report / Benchmarking

19 of 20 planted traps caught, zero unsourced numbers. A single prompt on the same documents caught none.



Metric	Target	Result
Trap recall	≥ 16 / 20	19 / 20
Unsourced numbers	0	0 (baseline 6)
Objections on clean case A	≤ 20%	0
Extraction row accuracy	≥ 97%	100%
Evidence-link accuracy	≥ 95%	100%
Time per case	≤ 180 s	4.5 s
Cost per case	≤ USD 0.50	USD 0.00

How measured. Run of 2026-10-14 in mock mode: full pipeline plus the deterministic Challenger battery, no Gemini calls, 3 synthetic businesses, 43 documents, answer key traps.json. Baseline is a stub in this mode; a live Gemini 3.8 Flash run replaces both before submission.

Additional details / Future development

Where this goes next, and what we do not claim yet

Pilot Q1 2027

One SEA bank or MFI, real consented SME files; underwriter decisions replace synthetic traps

→

Integrate Q2–Q3 2027

Open-banking feeds (Brankas, Finverse) beside PDFs; hooks into loan origination systems

→

Scale 2028

Same pattern for trade finance and micro-insurance; shared Challenger check library

What we do not claim yet, and which phase answers it

Synthetic benchmark

Mock-mode numbers measure the pipeline and tools. The pilot replaces them with real-data results.

Forged documents

Hash and balance-chain checks catch crude edits. Open-banking data closes the gap for the rest.

Platform sellers

Shopee and Grab already lend to their own sellers. We target banks and MFIs lending outside platforms.

positioning, not a feature

Humans decide, every figure and decision is traceable, the prototype uses watermarked synthetic data only, and production keeps documents and model endpoints in-region.

Provide links to your:

GitHub public repository

github.com/<team>/ledgerlens

English README, architecture, run instructions, synthetic-data notice

Demo video (3 minutes)

YouTube public link

Problem in 20 seconds, click-to-source and Challenger scenes, benchmark at 2:20

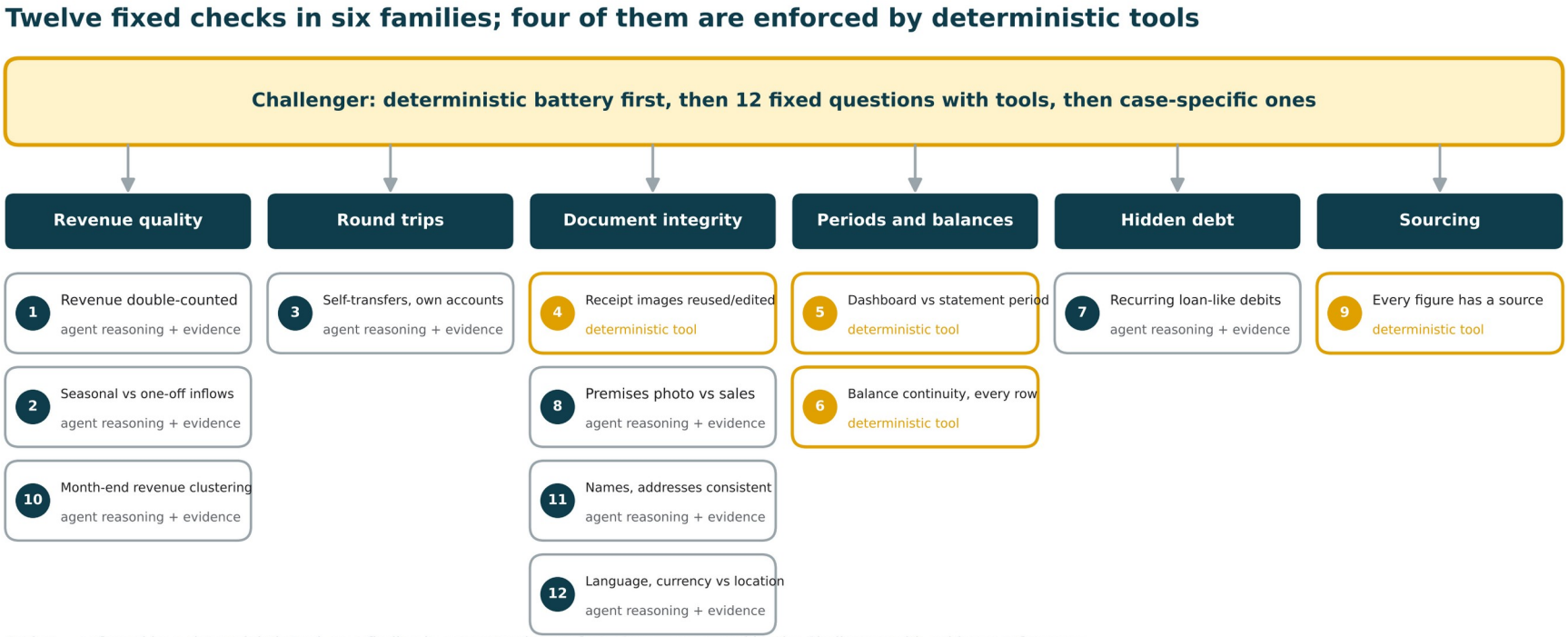
Final product

ledgerlens-<hash>.a.run.app

Cloud Run service, three demo cases preloaded, warm through evaluation

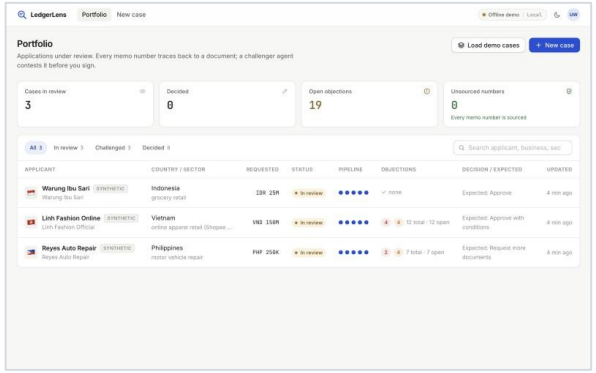
Links are inserted at submission on 18 October 2026.

Appendix: the Challenger's 12 fixed checks



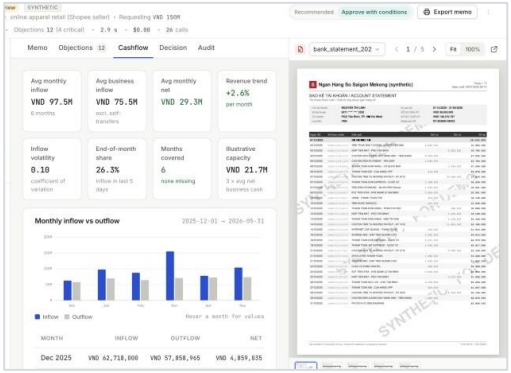
Amber = enforced by a deterministic tool, so a finding is guaranteed to surface. Grey = reasoned by the Challenger with evidence references.

Appendix: portfolio, cash flow and audit views



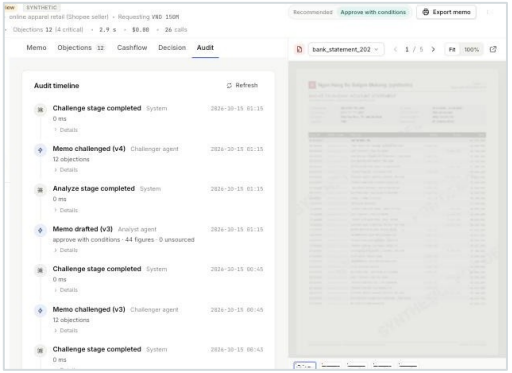
Portfolio

Cases in review, open objections and unsourced numbers across the book; filters by status.



Cash flow

Deterministic monthly model: inflow, net, trend, volatility, end-of-month share, repayment capacity.



Audit timeline

Every stage and agent call with version, model and prompt; the trail a regulator can read.

Also in the build: dark and light themes, PDF text-layer cross-check on every extracted amount, content-hash LLM cache, cache_only demo mode, budget cap, printable paper memo with footnoted sources.

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Build in JAPAC.
Build for the world.

Thank you

